TATRASI AMAR

Software Engineer | Full-Stack Developer (Java · Spring Boot · React · Next.js)
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SUMMARY

Full-stack software engineer building scalable multi-tenant enterprise systems and real-time IoT platforms with Java, Spring Boot, React, TypeScript, Next.js and PostgreSQL. Built in-house tools that replaced commercial vendor software, saving ₹3–4 Lakhs a year in licensing. Experienced in MQTT telemetry, Quartz scheduling, Razorpay payments and AI agent workflows.

TECHNICAL SKILLS

- **Frontend:** React, Next.js, TypeScript, JavaScript, Tailwind CSS, Shadcn UI, Recharts, HTML5, CSS3
- **Languages & Backend:** Java, Spring Boot, Spring Data JPA, Hibernate, REST APIs, Python
- **Databases & ORM:** PostgreSQL, Supabase, MySQL, Drizzle ORM, Row-Level Security (RLS)
- **Real-Time & Distributed:** MQTT (Paho/HiveMQ), Quartz Scheduler, WebSockets, Modbus, SCADA/HMI
- **Cloud, AI & DevOps:** Google Gemini LLM API, Docker, Git, GitHub, Postman, Selenium, CI/CD, Razorpay API, Agile/Scrum

PROFESSIONAL EXPERIENCE

Software Engineer (SE-1) | Terracos Technologies Pvt Ltd, Hyderabad | **Jul 2025 – Present**

- Engineered full-stack multi-tenant platforms (HRMS, Payroll, Industrial IoT BMS) with React, TypeScript, Spring Boot and PostgreSQL for 1,000+ active users.
- Architected an in-house HRMS and payroll engine with configurable salary structures, tax formulas and bank IFSC validation (100% calculation accuracy), saving ₹3–4 Lakhs a year in SaaS licensing.
- Implemented asynchronous MQTT ingestion for telemetry from 10+ industrial sensors, cutting end-to-end data latency by 50%.
- Integrated Quartz Scheduler for background jobs, daily attendance aggregation and scheduled PDF/email dispatch.
- Built Razorpay payment flows with webhook handling for transaction verification, automated billing and invoicing.
- Designed role-based (RBAC) React, TypeScript and Tailwind interfaces, standardizing reusable UI components across 3 client platforms.
- Applied SEO and Static Site Generation (SSG) to React apps, reducing page load times by 40%; built Selenium test workflows in Agile/Scrum sprints.

Software Engineer Intern | Terracos Technologies Pvt Ltd, Hyderabad | **Apr 2025 – Jun 2025**

- Developed responsive BMS dashboards for HVAC metrics, power distribution and fault alarms with Recharts and WebSockets.
- Contributed to PostgreSQL/Supabase schema design for the Attendance Management System, supporting real-time audit logs.
- Resolved frontend state management and API integration bottlenecks, improving dashboard rendering speed.

KEY PROJECTS

Terracos AI – Agentic AI B2B Platform *(Next.js 14, Google Gemini LLM, Drizzle ORM)*

- Built an AI prospecting platform on the Next.js 14 App Router and Gemini API that autonomously discovers, scores and qualifies business leads.
- Built server actions, optimized PostgreSQL persistence with Drizzle ORM, and added transactional email via the Resend API.

Enterprise HRMS & Payroll System *(Spring Boot, React, TypeScript, PostgreSQL)*

- Built biometric attendance, leave tracking and dynamic payroll calculation for 1,000+ employees.
- Used a Controller-Service-Repository architecture with structured exception handling and Quartz cron jobs for reporting.

BMS & Smart IoT Telemetry Platform *(Java, Spring Boot, MQTT, React, Recharts)*

- Built a live HVAC and power-meter telemetry pipeline with threshold-breach alerts, auto-reconnection and historical data export.

EDUCATION

B.Tech, Computer Science & Engineering | Jayaprakash Narayan College of Engineering | **2021 – 2024**

Diploma, Electrical & Electronics Engineering | Jayaprakash Narayan College of Engineering | **2018 – 2021**

CERTIFICATIONS

Full Stack Development with Cloud Computing – EY-GDS | Data Structures and Algorithms in Java – GeeksforGeeks | Java Fundamentals – Oracle Academy | Cloud Computing – IBM | CCNA – Cisco | Claude 101 – Anthropic